# Backtesting Probability of Default Models

## Frequentist Coverage of the Jeffreys Test

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# Probability of Default Backtesting

- As risk quantification is a regulatory requirement, practitioners pay significant attention to complying with these requirements.
- In addition to following the prescribed risk quantification process, practitioners must assess the appropriateness of model estimates. This is typically referred to as backtesting.
- The primary purpose of backtesting in this context is to ensure that model estimates accurately predict the occurrence of defaults, i.e., that Probability of Default (PD) estimates provide reliable forecasts of default rates.
- In practice, backtesting is one of the most comprehensive evaluation and validation exercises, typically investigating model estimates across different portfolio subsegments and time dimensions.
- Backtesting often includes a set of quantitative and qualitative tools used to draw an overall conclusion on the appropriateness of PD model estimates.
- One important aspect of the quantitative tools is the statistical testing framework commonly used for backtesting.

# Statistical Tests

- As part of the quantitative testing framework for backtesting, practitioners typically use the following three tests:
  - exact binomial test;
  - z-score test;
  - Jeffreys test.
- Given the conservative nature of IRB modelling, hypothesis tests are typically formulated to assess whether there is evidence of possible underestimation.
- In simple terms, testing for underestimation means that practitioners employ one-sided versions of the tests. Nevertheless, certain evaluation and validation aspects also require testing for accuracy or precision, which involves two-sided testing.
- The above-mentioned tests are typically used as alternatives for underestimation testing. However, practitioners often overlook that these tests rely on different underlying testing assumptions.
- While the Jeffreys test follows a Bayesian approach, the other two tests follow a frequentist approach. Further details on the differences in the underlying testing mechanisms are available here.
- The purpose of this presentation is to demonstrate the frequentist coverage of the Jeffreys test, justifying the consideration of all three tests for PD backtesting. Moreover, the simulation design shows that the Jeffreys test behaves similarly to the other two tests when applied to homogeneous and heterogeneous portfolios.

# The Jeffreys Test

For PD backtesting, the Jeffreys test is typically used to assess the evidence of PD underestimation. The p-value of the Jeffreys test under this assumption can be calculated as:

$$Pr(X \leq d) = \beta(q = PD, shape1 = nd + 0.5, shape2 = no - nd + 0.5)$$

where:

- $\beta$ is the cumulative distribution function of the beta distribution;
- $PD$ is the calibrated PD;
- $nd$ is the number of defaults;
- $no$ is the number of observations.

It should be noted that, although the Jeffreys test is commonly used to test for PD underestimation, it can also be used to assess the accuracy or precision of PD estimates. In other words, a two-sided version of the test can be used instead of the one-sided version.

# Simulation Study

The purpose of this simulation is to investigate the frequentist coverage of the Jeffreys test and demonstrate its suitability for PD backtesting.

The simulation is conducted under two assumptions:

1. homogeneous portfolio;
2. heterogeneous portfolio.

Both simulation scenarios rely on Monte Carlo simulations, in which defaults are independently generated from a binomial distribution. However, the simulation design differs depending on the assumed parameters of the binomial distribution.

For a homogeneous portfolio, the main assumption is that the overall portfolio PD represents the risk level of each exposure in the portfolio. The sample size is therefore defined as the total number of exposures in the portfolio.

For a heterogeneous portfolio, the main assumption is that the portfolio consists of exposures with different risk levels, represented by the PDs and corresponding sample sizes of individual rating grades.

Although the underlying simulation mechanism is the same, the implementation differs between the two scenarios. For a homogeneous portfolio, defaults are simulated for the entire portfolio at once. For a heterogeneous portfolio, defaults are first simulated independently at the rating-grade level and then aggregated to obtain the total number of defaults at the overall portfolio level.

# Simulation Study cont.

The simulation is based on the following rating scale:

```
##   rg    n    pd
##  RG1  300 0.005
##  RG2 1000 0.010
##  RG3 2200 0.025
##  RG4 3000 0.040
##  RG5 1800 0.070
##  RG6 1200 0.200
##  RG7  500 0.350
```

Under the homogeneous portfolio assumption, the overall portfolio `PD` is set to `0.07275`, while the sample size `N` is defined as the total number of exposures across all rating grades and set to `10,000`.

# Simulation Study cont.

The following steps outline the simulation design under the heterogeneous portfolio assumption:

1. for each rating grade, simulate the number of defaults from the binomial distribution using the corresponding `pd` and sample size `n`;
2. aggregate the simulated defaults across all rating grades (`ND`) and calculate the overall portfolio default rate;
3. given the simulated portfolio-level number of defaults `ND`, the overall portfolio size `N`, and a significance level $\alpha = 5\%$, calculate the lower and upper bounds of the 95% confidence interval using the Jeffreys test. The beta distribution shape parameters are set to `ND + 0.5` and `N - ND + 0.5`, respectively.;
4. create a coverage indicator that takes the value of `1` if the simulated default rate falls within the 95% confidence interval and `0` otherwise.;
5. repeat steps 1–4 a total of `100,000` times, storing the coverage indicator for each simulation;
6. calculate the frequentist coverage of the Jeffreys test as the average of the coverage indicators.

Under the homogeneous portfolio assumption, the simulation follows essentially the same steps, except that in step 1, the number of defaults is simulated directly at the portfolio level using the overall portfolio `PD` and sample size `N`.

# Simulation Results

